PANTRYPAL · PRODUCT DISCOVERY WITH AI AGENTS

Do agents beat one good prompt?

A multi-agent system that writes a cited discovery brief, measured against a single prompt on quality, cost and time.

[Your name] · [Course] · October 2026

The question

Where does a team of AI agents write a better discovery brief than a single prompt, and where does it only add cost?

Same input

10 customer interviews, 7 competitor rows, 9 market statistics, 1 idea brief

Same task

An 8-section discovery brief in which every claim cites its source

Same model

gemini-3.8-flash for every system, each run 3 times

First, the data: it is full of traps

9 of 10

questions in interview P06 are leading, and its answers simply agree

25

answer sentences repeated word for word across interviews

4 of 9

market statistics can't back a claim: no source, a blog, or from 2016

2 of 7

competitor rows were last verified more than a year ago

The EDA produced 19 findings, each ending in a rule: *"so the agent will..."*. They reach the agents as one file, checked against the data's fingerprints.

How the findings shaped the system

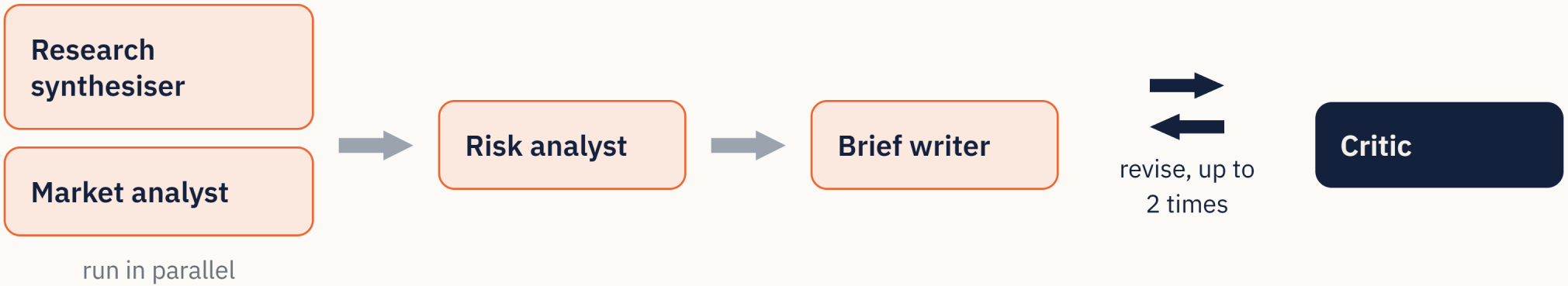
What the EDA found	What the system does about it
P06 asks leading questions	Evidence weight 0; search never returns it
25 sentences copied across interviews	Support counted by interviews in their own words; copied sentences flagged in every passage
S06-S09 can't back a claim	The citation check blocks them unless the line calls them unreliable
Models can invent quotes and figures	Every quote, date and number is checked against the data by code
Interview text could carry instructions	All data reaches the model as quoted sources, never as instructions

Five agents, each with its own inputs

Agent	Reads	Hands over
Research synthesiser	The interviews, through search and read tools	Segments, problems and disagreements, with quotes
Market analyst	Competitor rows and market statistics; no interviews	Threats, usable figures, checks of the idea's claims
Risk analyst	The two analyses only; no raw data	Assumptions ranked by risk, experiments
Brief writer	All three analyses	The 8-section brief, every claim cited
Critic	Each draft, the code check, the sources themselves	Objections with a fix; approve or revise

Every hand-off is a typed schema the model API enforces, so a malformed hand-off fails at once instead of drifting downstream.

A fixed pipeline with one review loop



Why a pipeline, not a supervisor

The order never changes, so plain code runs it. An AI supervisor would re-decide the same order every run, adding cost and variance.

When the loop stops

The critic approves and the code check passes; or 2 revisions are used; or a revision fails to reduce serious objections.

How each brief was scored

CODE

Citation checks

Sources exist; quotes, dates and numbers match the data

JUDGE + CODE

12 traps

Issues the EDA found. The judge must quote the brief; code verifies the quote

JUDGE + CODE

Faithfulness

Claims that misrepresent the source they cite, each one verified

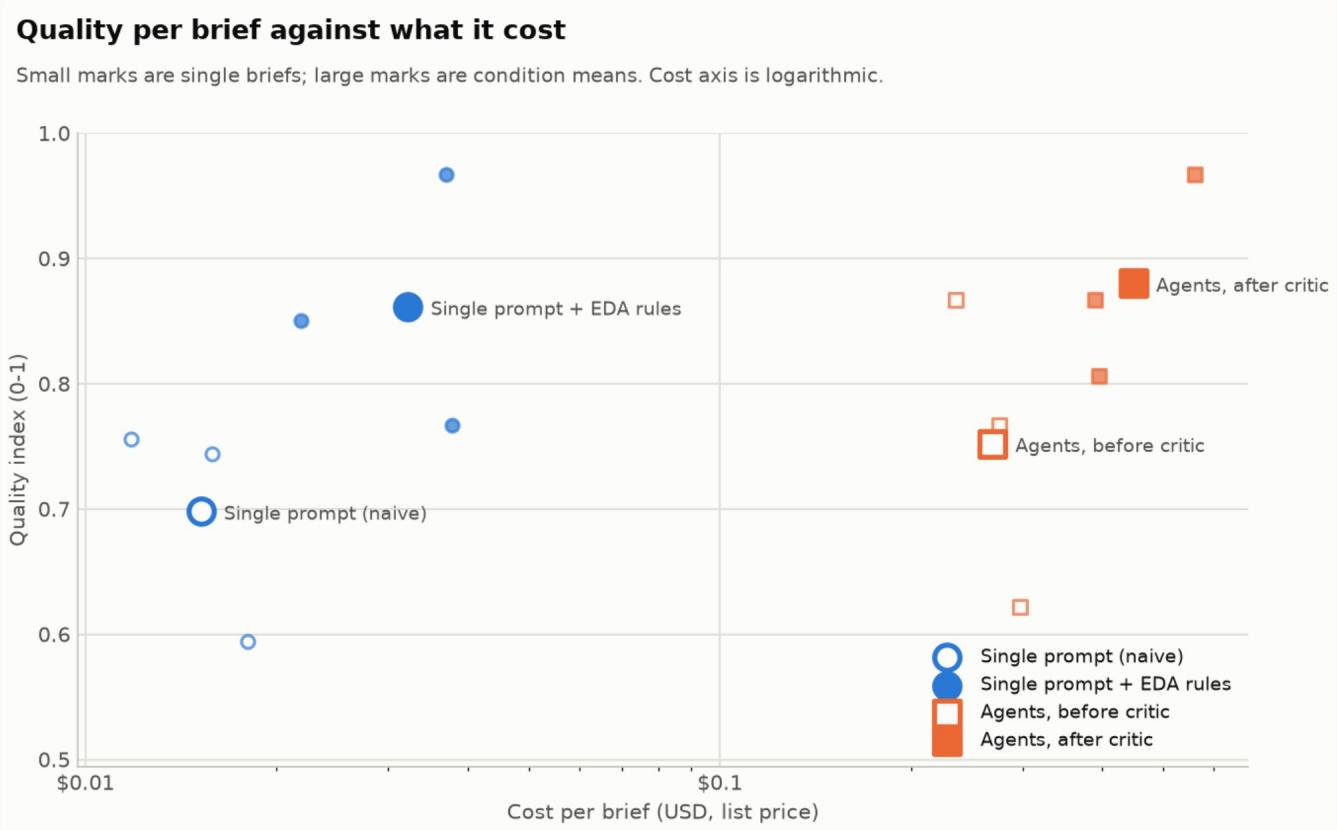
JUDGE

5 criteria

Completeness, grounding, calibration, actionability, clarity: 1 to 5

The judge, gemini-3.1-pro-preview, never learns which system wrote a brief. **Quality index** = the mean of traps handled, faithfulness and criteria, each from 0 to 1.

Results: three runs of each system



Single prompt
0.70 · $0.015 · 25 s

Single prompt + EDA rules
0.86 · $0.032 · 49 s

Agents, before critic
0.75 · $0.27 · 5.5 min

Agents, after critic
0.88 · $0.45 · 8.1 min

Quality · cost per brief · time. Ranges overlap: 0.77-0.97 for one prompt with rules, 0.81-0.97 for agents.

PantryPal discovery · 8 · gemini-3.8-flash, 3 runs each · Source: outputs/eval/eval_report.md

What each part adds

+0.16

The EDA's rules, in one prompt

Traps handled 36% → 92%, for 2 cents more and 24 seconds

−0.11

Splitting the work across agents

Misrepresented sources per brief 0.67 → 2.0: the writer only sees summaries. +$0.24, +4.6 min

+0.13

The critic loop

Catches those misrepresentations: 2.0 → 0.67 per brief. +$0.18, +2.7 min

The gain comes from knowledge and checking. Splitting the work mostly creates errors for the critic to fix.

Where the agents looped, drifted and missed

Looped

The critic checked sources one at a time and ran out of its 12 tool rounds: 1 of 3 runs failed. Fix: a forced verdict at the limit and batched lookups. Next 9 runs: 0 failed.

Drifted

The writer called S07 "73% of households abandon meal-planning apps within two weeks". S07 says "Most families struggle with dinner decisions". The critic caught it.

Missed by the critic

A brief said PlanPlate was verified on 2004-11-05; the data says 2024. The critic approved it. Only the code check caught the date.

Over-strict

The critic called experiment targets "invented numbers". New rule: a target is a proposal, marked [UNSUPPORTED], not a finding.

The brief: test further before building

The problems are real

Food goes off, households don't coordinate their shopping, and deciding what to cook is a daily chore.

The product rests on an unproven habit

Keeping a pantry inventory up to date. Five interviews say they would quit if they had to type it in (one repeated sentence).

Willingness to pay is low

A few euros, and only if it clearly saves money; the market median is €3.50 a month.

Tests, in priority order

1. Low-effort inventory: 20 households, photo or barcode capture, 7 days
2. A weekly plan against a same-day dinner suggestion: 14 days
3. Price at €3.50 and €4.99 a month: landing-page test
4. A shared list against duplicate buying: 16 households

All 22 Gemini briefs, from every system, recommend: test further.

The answer to the central question

Worth the cost

- The analysis rules in the prompt: +0.16 for 2 cents
- A critic that checks claims against the sources
- Code checks for quotes, dates and citations

Only added cost

- Splitting one task across five agents: more source errors
- 14× the cost and 10× the time for about the same quality

For a task that fits one context: one well-informed prompt, code checks and a critic pass. The next test is exactly that combination.

Limits: 3 runs each and the ranges overlap · a Gemini judge graded Gemini briefs; human scoring still to do · synthetic data · a scoring bug was found and fixed, and every brief was re-scored.